

Day 9 task: 4 Key AI Agent Architectures

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1 Introduction

When building complex AI agent systems, a single agent may not be the most effective solution. Instead, different "agentic frameworks" or architectures can be used to manage complex tasks. This report details four distinct architectures for building multi-agent systems, each with specific advantages.

2 The Four Agent Architectures

2.1 1. Prompt Chaining

- **What it is:** A linear, sequential architecture where the output of one agent is passed directly as the input to the next agent. The process flows in a single direction (Agent 1 → Agent 2 → Agent 3).
- **Example:** A blog-writing system where Agent 1 writes an outline, Agent 2 evaluates and refines the outline, and Agent 3 writes the full blog post based on the refined outline.
- **Advantages:**
 - **Improved Quality:** Each agent focuses on a single, specialized task, leading to a more accurate and higher-quality final output.
 - **Greater Control:** You can refine the system prompt and model for each specific step in the chain.
 - **Specialization:** Allows you to use different AI models for different steps (e.g., a fast, cheap model for outlining and a powerful, expensive model for the final content generation).
 - **Easier Debugging:** The linear flow makes it simple to identify which step in the process is failing or producing poor results.

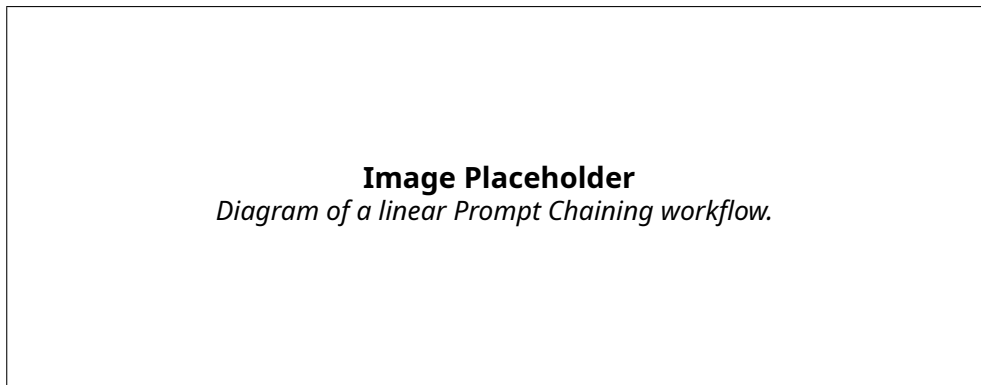


Figure 1: Prompt Chaining Architecture

2.2 2. Routing

- **What it is:** An architecture where an initial "router" or "classifier" agent analyzes the user's input and routes the task to one of several specialized agents based on its nature.
- **Example:** An email processing system. An initial agent classifies an incoming email as "Sales," "Support," or "Finance." It then routes the email to the specialized "Sales Agent," "Support Agent," or "Finance Agent," each with its own persona and tools.
- **Advantages:**
 - **Optimized Responses:** Allows for different personas, tones, and specialized knowledge for each route (e.g., the Support agent is empathetic, the Finance agent is formal).
 - **Scalable and Modular:** You can easily add new routes (e.g., a "Marketing Agent") without breaking the existing ones.
 - **Efficient:** The router agent is often a fast, cheap model, and the specialized agents only run when needed.
 - **Human Escalation:** A route can be configured to escalate a task to a human (e.g., a "High Priority" route sends a notification instead of an automated reply).

2.3 3. Parallelization

- **What it is:** An architecture where the user's input is sent to multiple agents *at the same time*. The outputs from all agents are then collected, merged, or aggregated by a final agent to create a comprehensive response.
- **Example:** Analyzing a piece of text. The text is sent to an "Emotion Agent," an "Intent Agent," and a "Bias Agent" simultaneously. A final "Report Agent" takes the analyses from all three and combines them into a single report.
- **Advantages:**
 - **Faster Analysis (Lower Latency):** All analyses run in parallel, which is much faster than running them in a chain.
 - **Comprehensive Review:** Gathers multiple perspectives on the same data.
 - **Specialization:** Each parallel agent can have a highly specific prompt. You can even use different models in parallel (e.g., send the same prompt to Claude, Gemini, and GPT) and merge their "best" insights.

2.4 4. Evaluator / Optimizer

- **What it is:** A cyclical architecture that creates a feedback loop. An initial agent generates content. An "Evaluator Agent" then checks this content against a strict set of criteria.
 - If the content **passes**, it is approved and sent to the final output.
 - If the content **fails**, it is sent to an "Optimizer Agent" along with the evaluator's feedback. The optimizer revises the content, and the new version is sent **back** to the evaluator. This loop repeats until the criteria are met.
- **Example:** A biography writer. Agent 1 writes a biography. The Evaluator checks: "Does it include a quote? Is it humorous? Are there no emojis?" If not, it sends feedback (e.g., "Missing quote"). The Optimizer adds a quote and resubmits.
- **Advantages:**
 - **Ensures High Quality:** The output **must** meet a defined standard before it's released, guaranteeing quality and consistency.
 - **Reduces Manual Review:** Automates the "review and revise" cycle that a human would normally perform.
 - **Iterative Optimization:** The agent system continuously improves its own output until it is correct.